

VILNA AGCM, AB, Canada

WMO: 717610

Lat: 54.130N

Lon: 111.921W

Elev: 2116

StdP: 13.61

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-27.6	-21.0	-34.0	0.9	-27.3	-27.0	1.3	-20.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	81.4	62.8	78.3	61.8	75.4	60.7	66.3	76.9	64.2	74.6	62.4	72.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	90.0	72.6	60.0	83.7	69.6	58.3	78.6	67.2	32.2	76.6	30.5	74.5	29.1	72.6	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-36.6	87.1	5.5	1.9	-40.6	88.5	-43.8	89.6	-46.9	90.7	-51.0	92.1
			WB	-36.8	70.7	5.5	2.7	-40.8	72.6	-44.0	74.2	-47.1	75.7	-51.1	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	10.1	9.5	21.7	37.0	50.6	57.9	61.6	59.5	50.9	38.1	21.6	7.9		
	DBStd	22.49	15.69	14.26	13.84	9.24	8.19	5.46	4.58	5.51	7.92	8.41	12.29	14.81		
	HDD50	6371	1238	1135	878	399	93	9	0	3	82	379	851	1304		
	HDD65	10745	1703	1555	1343	841	447	220	125	182	424	835	1301	1769		
	CDD50	1143	0	0	0	8	112	245	361	298	110	9	0	0		
	CDD65	43	0	0	0	0	2	6	21	12	2	0	0	0		
	CDH74	927	0	0	0	2	108	145	310	241	120	1	0	0		
	CDH80	142	0	0	0	0	17	17	48	34	26	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	17.2	0.7	0.6	0.7	1.2	1.7	3.0	3.3	2.4	1.4	0.9	0.8	0.7		
(15) Precipitation	PrecMax	20.7	2.0	1.3	1.4	3.4	3.2	6.4	7.2	5.4	3.6	2.2	2.2	1.4		
	PrecMin	10.9	0.0	0.0	0.2	0.1	0.2	0.9	1.3	0.9	0.1	0.2	0.1	0.0		
	PrecStd	2.6	0.4	0.3	0.4	0.7	0.8	1.3	1.1	1.0	0.9	0.4	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	42.3	52.4	71.8	82.5	82.4	85.0	84.1	84.2	70.4	52.9	41.4		
		MCWB	37.5	36.1	42.3	50.3	57.1	61.7	67.2	65.0	60.3	54.1	42.7	34.7		
	2%	DB	39.4	37.6	46.2	63.5	77.0	78.1	80.7	79.6	77.1	61.1	45.0	36.2		
		MCWB	34.6	33.2	38.2	46.2	54.7	60.1	65.3	64.0	58.6	49.0	38.9	31.6		
	5%	DB	36.1	33.9	42.0	57.8	72.1	74.6	77.4	76.5	72.0	56.1	40.3	32.3		
		MCWB	32.5	30.8	36.0	43.7	52.7	58.5	64.0	63.1	56.8	45.7	36.0	29.1		
	10%	DB	32.2	30.7	38.5	52.2	68.1	71.2	74.4	73.4	66.6	51.4	36.4	27.9		
		MCWB	29.5	28.2	33.7	40.8	50.6	57.4	62.0	61.3	54.3	43.2	33.3	25.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	37.0	43.4	51.3	58.9	65.5	70.2	68.8	62.9	54.6	43.9	35.5		
		MCDB	43.0	41.2	51.2	71.0	76.3	77.6	80.0	79.1	79.2	67.2	50.5	40.2		
	2%	WB	35.1	33.7	38.8	46.8	56.6	62.4	67.1	66.1	59.9	50.0	39.5	31.9		
		MCDB	38.8	36.9	45.2	61.9	71.7	73.0	77.7	75.9	74.1	59.0	44.3	35.8		
	5%	WB	32.5	31.3	36.3	44.3	54.4	60.5	64.9	63.8	57.5	46.9	36.6	29.0		
		MCDB	35.9	33.9	41.4	56.2	69.6	70.9	75.2	73.8	70.9	54.3	40.1	32.0		
	10%	WB	29.4	28.0	34.0	41.7	52.2	58.7	63.0	62.0	55.2	44.2	33.3	25.8		
		MCDB	32.3	30.8	38.3	50.9	65.0	68.1	72.4	71.7	65.2	50.8	36.1	27.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	19.0	17.8	19.4	24.4	21.3	21.6	22.7	23.2	18.4	15.1	15.5		
		MCDBR	17.9	20.6	20.4	29.2	31.5	28.2	27.3	28.4	33.7	26.9	19.2	19.0		
		MCWBR	14.2	16.8	13.8	15.6	14.1	13.1	14.4	14.8	16.9	15.8	13.3	15.4		
	5% WB	MCDBR	17.8	19.7	19.6	27.1	28.7	23.6	25.5	25.9	31.8	25.3	18.2	18.5		
		MCWBR	14.3	16.3	13.4	15.0	13.2	12.2	14.0	14.3	16.4	15.4	13.0	15.4		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.264	0.297	0.327	0.343	0.356	0.357	0.365	0.308	0.278	0.250	0.235		
	taud		2.296	2.315	2.331	2.380	2.389	2.370	2.407	2.341	2.508	2.562	2.450	2.296		
	Ebn at Noon		229	268	281	286	286	283	280	270	277	264	236	214		
	Edh at Noon		18	24	30	33	35	36	34	35	25	19	15	14		
(44) All-Sky Solar Radiation	RadAvg		283	613	1096	1479	1812	1872	1907	1558	1087	613	288	204		
	RadStd		31	47	82	126	144	128	102	116	115	88	34	19		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page